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Contents

1 Chapter One: Introduction **9**

1.1 Background of the study 9

1.2 Problem Statement 10

1.3 Aim and objectives 10

1.3.1 Aim 10

1.3.2 Objectives 11

1.4 Research Questions 11

1.5 Significance of study 11

1.6 Summary 11

2 Chapter Two: Literature Review **12**

2.1 Key Concepts and Terms 12

2.1.1 Artificial Intelligence 12

2.1.2 Tuberculosis Detection 12

2.1.3 Explainable AI 12

2.2 Current models and Weaknesses 13

2.2.1 CAD4TB 13

2.2.2 Lunit insight CXR 13

2.2.3 QureAI 14

2.3 Research gaps 14

2.4 Summary 14

3 Chapter Three: Methodology **16**

3.1 Data Collection 16

3.1.1 Data collection techniques 16

3.1.2 Chest X-ray images 16

3.1.3 Radiologist and clinician reports 17

3.1.4 Sampling 17

3.2 Machine learning approach 18

3.2.1 Requirements Definition 19

3.2.2 Data curation 19

3.2.3 Model creation 19

3.2.4 Model Training 19
3.2.5 Explainability integration 19
3.2.6 Model testing 20
3.2.7 Model deployment 20
3.3 Application wrapper approach 20
3.4 Ethical consideration 20
3.5 Summary 21

4 Chapter Four: Limitation of Study 22
4.1 Training Hardware 22
4.2 Explainable engine choice 22
4.3 Summary 22

5 Chapter Five: Research Plan 23
5.1 Time Frame 23
5.2 Budget 23

1 Chapter One: Introduction

This chapter introduces the study project, presenting the background of tuberculosis and how Machine learning imaging models together with explainability can take part in its management. The chapter focuses on the research problem, the aim of the study and all objectives guiding the research. The chapter also outlines the research questions together with their supporting questions. The main goal of this chapter is to provide a comprehensive overview of the project and why it is important to develop the tuberculosis detection model with explainable visualizations.

1.1 Background of the study

Tuberculosis, also known as white death, is a contagious disease caused by Mycobacterium tuberculosis bacteria [Hawn et al., 2014]. According to the World Health Organization, tuberculosis has caused over 2.5 million deaths in the African region.In 2021, the incidence of tuberculosis in Malawi was 132 per 100,000 population [WHO, 2023]. In the past few years, the country has managed to combat tuberculosis leading to 31% decline of TB incidence with the TB/HIV coinfection rate being dropping by 43% [KUHES, 2024]. To achieve the goal of 50% reduction in incidences by the ministry of health, the country imported devices like GeneXpert to aid in fighting against TB. The ministry of health to also release guides to train healthworkers on how to fight against the disease in a modern way with these devices. Apart from GeneXpert, the ministry of health also imported a collection of other devices like digital X-ray and CAD4TB. Computer Aided Detection for Tuberculosis commonly known as CAD4TB is a portable Artificial Intelligence(AI) tuberculosis detection device [KUHES, 2024]. The CAD4TB uses chest X-ray images and an AI model to detect tuberculosis. This device has predictive analysis and can work locally without internet connection. Around seven mobile clinics with CAD4TB devices deployed is in malawi and they managed to screen around 395,279 people [Delft-Imaging]. As this adoption of the imaging models grows, trusting responses is crucial for both doctors and clinicians [Fehr et al., 2021]. The models used commonly in blantyre are the CAD4TB and QureAI [Twabi et al., 2021]. Trusting the responses from models has been done in a “black box” form, where users do not know the internal working of the models to lead to the presented response [Ribeiro et al., 2016].

In a review, both CAD4TBv5 and Qure.ai qXR v2.0 lack the component of explainability. Explainability helps in fault detection by the clinicians, for example, in a TB image the models response will be returned with highlights of the patches as soon as the patches are in the wrong position clin-

ician can mark that response as invalid [Müller et al., 2023]. Secondly, Malawi as a country has a large shortage of radiologists, as these AI devices are coming in they are substituting radiologists in the hospitals that do not have radiologists [Chinkhumba, 2025]. The integration of explainability will be more critical in this period as models are being used by partially skilled clinicians. Addition of explainability will assist the clinicians ease in understanding the models response and be able to validate the response without a secondary source [Alharthi et al., 2024]. The lack of an explanation in the responses from the adopted models by the ministry of health presents a challenge as clinicians are not able to validate the responses without using a secondary resource like another model or a human expert [Delft-Imaging]. In a field of high stakes like healthcare it is important one trusts the model response whilst understanding the inner working as errors can cost lives [Müller et al., 2023]. The component of explainability in the current models like CAD4TB is by use of heatmaps which show where the model looked but not what the model concluded.

1.2 Problem Statement

Malawi is in a transition period of moving away from sputum testing to using AI imaging models for tuberculosis detection. In a study by the malawi liverpool wellcome trust, it was found that radiologists made errors in TB detection in one out of five times of cases while qureai had a 92% accuracy [Twabi et al., 2021]. This means that if radiologists together with AI models are being used, the errors will be reduced. QureAI is currently deployed at Queen elizabeth hospital among other hospitals and it demands that an operator with expertise to be available to validate the models response [Delft-Imaging]. This is a challenge as the country has a shortage of radiologists. By adding explainability to the models, it will be easier for the partially skilled clinicians to validate the models response without a secondary source. This will increase trust, reduce errors and increase adoption of the models. It will also reduce overreliance and automation bias as errors are now visible to the clinicians.Henceforth, there is a need a comprehensive tuberculosis model with explainable visualizations to aid the partially skilled clinicians easily catch false positives in the model and the unskilled clinicians to easily understand the model's response.

1.3 Aim and objectives

1.3.1 Aim

The aim of this project is to develop a tuberculosis detection model with explainable visualizations.

1.3.2 Objectives

The objectives of this project are:

- developing a dataset from available datasets (NIH) and locally sourced
- To finetune a tuberculosis detection model.
- To develop an explainable visualization for the tuberculosis detection model.
- To develop a multiplatform application for interaction with models and visualization engine.

1.4 Research Questions

The research questions of this project are:

- How does explainable AI impact clinicians in their decision making for AI tuberculosis detection?

1.5 Significance of study

The development of a tuberculosis model with explainability is of great importance, as it allows catching errors made by model during diagnosis. It also allows the clinicians to easily understand under the hood workings of the model hence better guide people responsible for training models on how to patch up models. This study is also significant as it directly combats tuberculosis whilst reducing cost and increasing adoptability parameters by partially skilled clinicians. As imaging models increase in adoption, catching errors in them has never become more important as these errors can lead to misdiagnosis and negatively affect the on patient health.

1.6 Summary

In conclusion, this study aims to develop a tuberculosis detection model with explainable visualizations. The development focuses on catching false positives and false negatives by using explainability engines. By providing a portable multiplatform interface it will have a similar or larger impact as CAD4TB. The portability and low-cost of the model will also increase adoption in local hospitals and assist in the combating of TB in Malawi.

2 Chapter Two: Literature Review

Chapter Introduction

This chapter presents the literature review for the development of a tuberculosis detection model with explainable visualizations, structuring the groundwork for the building the interface, imaging model and integrating of explainable visualizations engines. The chapter begins with discussion of key concepts so to unfold the meaning of terms used in the study then later evaluates current systems and their weaknesses. The chapter concludes by looking into the research gap and how this study will address the gap.

2.1 Key Concepts and Terms

2.1.1 Artificial Intelligence

Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making [Mazeika et al., 2025]. Artificial intelligence has multiple fields such as machine learning, deep learning and computer vision. Machine learning is a subset of AI that focuses on the development of algorithms that can learn from data and make predictions or decisions without being explicitly programmed [Ciresan et al., 2012]. Most imaging models are built with integration of deep learning and computer vision.

2.1.2 Tuberculosis Detection

Tuberculosis detection is the process of identifying the presence of tuberculosis in a patient. Tuberculosis detection can be done in numerous ways namely, skin test with tuberculin injection, blood tests, sputum tests where the patient mucus is tested for tuberculosis [Mayo-clinic]. Malawi has adopted most of these techniques with an addition of CAD4TB, digital X-ray and GeneXpert [KUHES, 2024]. Tuberculosis being the ninth leading cause of death it is important that early detection is achieved.

2.1.3 Explainable AI

This is a field of research that explores methods that provide humans with the ability of intellectual oversight over AI algorithms [Ribeiro et al., 2016]. Explainable AI aims to make AI decisions transparent, understandable, and interpretable. The lack of interpretability in AI systems has raised concerns about trust, accountability, and fairness. Without explanations user is left in the dark unable

to understand why the decision was made or what could be done for improvement [Alharthi et al., 2024]. Models might sometimes predict an outcome with high accuracy, but the reasoning behind it is unknown. The lack of explainability leads to trust and accountability issues, it also leads to easier debugging of models hence finding points of improvements.

2.2 Current models and Weaknesses

There are multiple AI integrated tuberculosis detection models already developed. The following are the most notable ones.

2.2.1 CAD4TB

Computer Aided Detection for Tuberculosis(CAD4TB) is an AI powered tuberculosis detection device. It is developed and owned Delft Imaging. It integrates AI chest x-ray software with EPICON's epi control platform [Delft-Imaging, 2025]. The AI software works by marking lung field with abnormalities as TB, assigning a score in percentage and producing a heatmap of chest abnormality. The Ultra-portable X-ray powered with CAD allows populations living in remote areas without internet, to access high quality TB screening services [CheckTB, 2025].

The main downside to CAD4TB is that it does not have concrete explainability not simply heatmaps. This is important as it assist the operator to understand the models response and be able to mark it as a false positive or a false negative. It will also assists in debugging of models so that they know areas of improvement when reports are being made. Secondly, the device is expensive compared to having your own model that is crossplatform.

2.2.2 Lunit insight CXR

The Lunit INSIGHT CXR is a deep learning software that assists radiologist in chest x-ray. The AI can detect multiple diseases namely; atelectasis, calcification, cardiomegaly, consolidation, fibrosis, mediastinal widening, nodule, pleural effusion, pneumoperitoneum, acute bone fracture, and tuberculosis. Like the CAD4TB, it also includes the heatmap analysis pane. It also has abnormality scores indicating the probability of the judgement [insight cxr4, 2025].

Unlike the CAD4TB, this has a small component of explainability as it shows the patches that make it think it has a lung disease but the issue is that these marks are general without any specificity to what disease each patch suspects and a score for each patch. Some disease share similar patch shapes hence the model might be confused. Furthermore, another downside to this is that it lacks

portability hence cannot be used in mobile for TB campaigns as it will need very strong devices which might be more expensive compared to CAD4TB. Lastly, the models cannot be trained and finetuned with the new data being collected by the hospital [insight cxr4, 2025].

2.2.3 QureAI

This is an AI model that is currently deployed here in Malawi at the queen elizabeth amongst other hospitals. It is developed by Qure.ai Technologies Private Limited in USA. The model is multipurpose as it works on detection of TB, heart failure, Stroke and lung cancer. It also has case management component where reports are sent via the qxr app. A user simply uploads the image and the model returns a response with a heatmap and a score [Twabi et al., 2021]. In a test by the Malawi liverpool wellcome trust, it had a 92% accuracy in TB detection.

Similar to CAD4TB, the core downside is lack of solid explainability. The heatmaps are not explainable as they do not show exactly what the model concluded if it was actually using those points marked.

2.3 Research gaps

Multiple contextual challenges exist which justify the need for building a local tuberculosis detection model with explainability visualizations. Most of the models like the QureAI has a component of explainability by use of heatmaps where after the models marks a chest X-ray image as having tuberculosis, it highlights the patches that it thinks are tuberculosis. This is just half of the equation as it does not exactly severely test the model by image segmentation and making perturbed dataset. It simply adds overlays to the images [Suthers, 2025]. Secondly these models lacking explainability component of explainability which risks clinicians in OneStopTB clinics from making errors in diagnosis. The research addresses the gap by proposing a tuberculosis detection model with explainable visualizations. The model will be portable and can be used on different type of devices. The models used in malawi using AI to detect presence of tuberculosis and then use heatmaps to show where the model looked. The heatmaps are not explainable as they do not show exactly what the model concluded if it was actually using those. This is a gap that the proposed system will fill.

2.4 Summary

This chapter has looked at the artificial intelligence, tuberculosis detection and explainability. The chapter also evaluated current models and their shortfalls. After evaluating these models, a gap was found that the proposed system will fill. The upcoming chapter will discuss the methodology of the study, pinpointing ways on how best the model will be developed.

3 Chapter Three: Methodology

Chapter Introduction

This chapter outlines the research methodology to be utilized in the development of the suggested tuberculosis model with explainability inclusivity. The sections looks into the research approach which emphasizes practical field observation with development techniques. The research strategy is also discussed in the chapter where a system design approach is chosen. The chapter also discusses the data collection and analysis techniques to be used in the study.

3.1 Data Collection

Different forms of data are to be for this study namely; chest X-ray images and radiologist and clinician reports. The data will be collected from multiple sources namely; the National Institute of Allergy and Infectious Diseases (NIAID), the Queens Elizabeth Central Hospital (QECH) or any TB clinic in Malawi and Kaggle.

3.1.1 Data collection techniques

For each of the data to be collected, different techniques will be used so that the data is safely acquired and its integrity is kept intact. Data integrity is important in this research as to prevent biases in conclusions made.

3.1.2 Chest X-ray images

The chest X-ray images will be collected from the NIAID and QECH. The images will be collected in a digital format and will be stored in a secure location. The images will be anonymized using UUIDS to prevent identification of the patients. The images will be collected in a way that the data is not biased towards any particular group.

NIAID Data will be collected from the NIAID by using desk research approach where one asks for data from an already existing data source.The image data from NIAID will be collected by applying for their portal access programme. Once accepted, the dataset is made available by use of the NIAID portal and it can be downloaded and used.

QECH The data will be collected from the queen elizabeth hospital by Information request technique. This will involve approval from the school and the hospital. A formal letter request atleast 10

chest x-ray images to be used for testing of the model. This will help to see how the model performs with data that is in-sample.

Kaggle Data will be collected from Kaggle by a technique called archival research where one will look for already existing datasets and use them. Considering how much open data is available on kaggle this technique is the best and has the least amount of friction. The data will be collected in a way that the data is not biased towards any particular group.

3.1.3 Radiologist and clinician reports

Radiologist and clinician reports are important to evaluate how they use the model and how they interpret the results. The reports will be collected from the QECH. The reports will be collected in a digital format. The reports will be anonymized using UUIDS on names to prevent identification of the radiologists. An academic visit to QECH is to be scheduled to visit the QECH radiology lab.During this visit the researchers will guide the researchers on how to use the model. A semi-structured interview is to be conducted with the radiologists to understand how they use the model and how they interpret the results.

Interview During the educational visit to the radiology lab at QECH, the researchers will conduct a semi-structured interview with the radiologists. The interview will feature questions querying about explainability, model users and impact . This will allow a deep look into how explainability will impact the radiologist. The interview will be conducted in a way that the radiologists are not forced to participate. The interview will also help in determining on how best to design the user interface for the model so that a familiar experience is built.

Observation During the educational visit, the researcher will observe the radiologists using the model. The researcher will also observe how and when radiologists use explainability like heatmaps to make decisions. This will assist in validation of the importance of explainability.

3.1.4 Sampling

Considering we have multiple data sources, the sampling will be differently for each type of data.

Chest X-ray images Multiple sources exist for chest x-ray images so in order to achieve an unbiased representation of the data stratified sampling will be used [PennState]. The method divides a population into distinct, non-overlapping subgroups (strata) based on shared characteristics and then uses random sampling to select individuals from each group. This unbiased approach will ensure that the data is not biased towards any particular group and the model to be built is not biased.

Radiologist and clinician reports The radiologist reports will be collected from the QECH. The reports will be sampled using purposive sampling. Purposive sampling where participants are chosen from based on their involvement in tuberculosis. This will allow participants pool to be knowledgeable and have a positive overall impact to the project. Having knowledgeable participants will allow for a deeper understanding of the model and how explainability will be adopted in the whole workflow.

3.2 Machine learning approach

This research will follow the Machine learning approach for development of model and follow the prototype approach for development of the interface. Following the machine learning approach for model is a better choice compared to others because if focuses more on reproducibility and data integrity which is important in this research to easily trace back where model went wrong in case of errors in model. The prototype approach is a better choice compared to others because it focuses on the development of a prototype which is important in this research as it allows for testing of the model and interface with users. In the development of the model, we will follow the pyramid approach, which involve revisiting requirements at each stage of the process. This will ensure that the model is developed in a way that it meets the requirements of the users [Giunchiglia et al., 2023]. The Figure 1 below shows how the pyramid model works. It is composed of the following components: Requirements Definition, Data curation, Model creation, Model training, Model testing and model deployment.

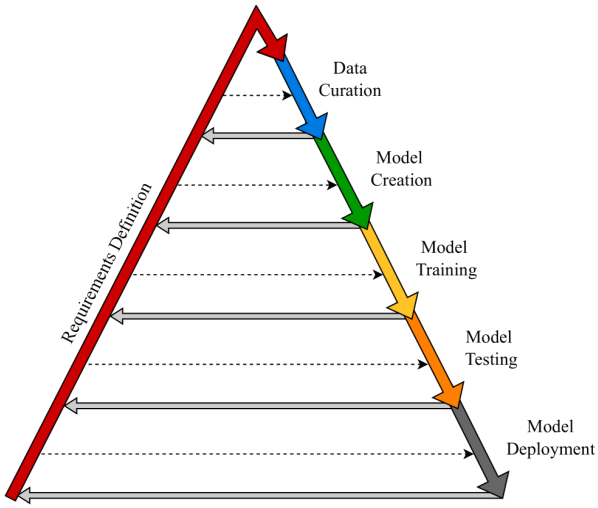

Figure 1: A figure of the pyramid approach for model development [Giunchiglia et al., 2023]

3.2.1 Requirements Definition

This involve specifying what each stage should do and making sure each stage achieves it. It means if in one stage the results gotten are not the right ones the researchers will go and re-evaluate the requirement and iterate.

3.2.2 Data curation

This involves the evaluation of the collected data. The collected data is later labeled as per the requirements. As for this study it will mean classifying the images in TB or not.

3.2.3 Model creation

This involves choosing of the base model for study. Since this will be development of an imaging model, the base model will either **YOLOV6** or **Detectro2**. The base model assists in the process of developing the whole model without the need to do everything from scratch. Considering the maintenance lifecycle YOLOV6 has it will be chosen [Li et al., 2022]. YOLOV6 is chosen because it is a convolution neural network and it performs more efficiently with object detection compared to other techniques like random forest. CNNs excel at object detection because of the weight sharing which help to detect tumors regardless of where the feature is selected [Li et al., 2022]. This will help as some patches of TB are small and might be missed by other models.

3.2.4 Model Training

This is an iterative process of feeding data into a machine learning algorithm to optimize its parameters, allowing it to learn patterns and make accurate predictions. This will reinforce the model to learn the patterns of TB and non TB images. Furthermore, the recording of losses made during the training so that one sees how to change the configurations of the base model in order to reduce the losses. Losses make the model perform poorly [Kumar et al., 2025].

3.2.5 Explainability integration

Once the model has been chosen, an explainability engine is will be chosen. Multiple explainability engines exist like the Local Interpretable Modal-agnostic explanations (LIME) and the SHapley Additive exPlanations (SHAP). LIME generates perturbations (variations) of a data instance to see how the model behaves and fits a simple, interpretable model around that local, specific prediction.

So for our model it will mean that it will work by perturbing the image and feeding the model into the trained model [Ribeiro et al., 2016]. SHAP takes a game-theoretic approach to explain the model response. SHAP assigns each feature an importance value for a particular prediction by calculating how much that feature contributes to the outcome, averaged over all possible combinations of other features [Lundberg and Lee, 2017].

3.2.6 Model testing

Each requirement made for the model will be tested to see if they have been met. If they have not been met it will demand that they have to be evaluated and see which stage affected this and go back to it. In this case, one of the tests will involve use of images from QECH which were not part of the training data. This will help to see how the model performs with data that is out of sample.

3.2.7 Model deployment

The model will need to be deployed on a local server since it is against the law to have medical data outside the country or premise. This will involve containerization of the model.

3.3 Application wrapper approach

Since the development will require an interface as a middle way interaction between the model and the clinicians. Prototype is best because the goal is to develop a quick version of the interface that is similar to the current interfaces used by the clinicians.

The Prototype approach will specifically the of incremental prototyping where the prototype is refined after every feedback. This will help not to develop something that the clinicians do not want. It will also allow for new ideas to be accepted which were not stated in the initial stage.

3.4 Ethical consideration

With artificial intelligence, ethics have to be considered because if one overlooks them social problems arise. The study will consider multiple of these ethical issues.

3.5 Summary

This chapter has looked at the research approach, strategy, sampling, data collection, data analysis and ethical considerations. The chapter has also looked at the research approach, strategy, sampling,

Risk	Description	Mitigation
Equipment bias	Images from high-end hospitals perform better	Test on diverse equipment/settings
Annotation bias	Radiologist labels reflect subjective judgment	Combine radiologist opinion + microl
Disease spectrum bias	Training on severe cases only	Include latent TB, pediatric TB, HIV-

Table 1: Ethical Considerations and Mitigations

data collection, data analysis and ethical considerations. The next chapter will discuss the system design and the development of the model.

4 Chapter Four: Limitation of Study

Chapter Introduction

This chapter discusses the limitations of the study. The limitations are discussed in terms of the scope of the study, the resources available, the time frame and the expertise of the researchers. The limitations are discussed in the context of the study and the research questions.

4.1 Training Hardware

The training of model is being conducted on a very low end device and this will affect the outcome of the model in terms of quality. This can be mitigated by training for longer periods but that can cause overfitting. Alternatively, the model can be trained on google collab which has better specs but due to free tier pricing limitations, the models can only be trained while connected to the internet and the resource sharing might lead to slow training time.

4.2 Explainable engine choice

The choice of explainable engine will heavily depend on how resource intensive each of them are and how much complexity they add to the output. Local Interpretable Modal-agnostic Explanations (LIME) is a good choice but it is not perfect as it is not being actively maintained and has not been updated to the latest builds of models. SHapley Additive exPlanations on the other hand is being currently explored and is uses both local and global agnostic approaches but this might make the application large hence losing on portability.

4.3 Summary

This chapter has looked at the limitations of the study. The next chapter will discuss the research plan.

5 Chapter Five: Research Plan

Chapter Introduction

This chapter outlines the research plan for the study. The research plan is divided into two main sections: the time frame and the budget. The time frame section outlines the activities to be undertaken and the duration of each activity. The budget section outlines the estimated cost of each activity.

5.1 Time Frame

ACTIVITY	DURATION
Proposal development	Month 1
Literature review	Month 1-2
Data collection	Month 2-3
System design and development	Month 3-4
Model training and testing	Month 4-5
System evaluation	Month 5
Report writing and submission	Month 6

Table 2: Research Time Frame

5.2 Budget

ITEM	ESTIMATED COST (MWK)
Data collection and transport	30,000
Internet and data access	50,000
Software tools and computing resources	60,000
Printing and documentation	40,000
Miscellaneous expenses	30,000
Total Estimated Cost	**210,000**

Table 3: Research Budget

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